Restoring Visibility Through Haze, Water, Glass and Plastic

A Comparative Study of Classical Image-Processing Methods, with the Dark Channel Prior (DCP) as the Baseline · CS7.404 Digital Image Processing, Monsoon 2026

Problem Statement

Camera visibility is degraded either by the medium between the scene and the sensor (atmospheric haze, turbid water) or by a protective or intervening element in front of the lens (fogged or dirty glass covers, translucent or aged plastic). These degradations differ physically, but all reduce contrast and visibility in similar ways. This project asks which classical, non-learning image-processing methods restore visibility best in each case, why they succeed or fail, and how far a synthetically generated degradation can stand in for a real capture. The Dark Channel Prior (DCP) is used throughout as the reference method and as the lens for explaining results. No machine learning is used in any restoration or degradation step.

Objectives

- Implement the Dark Channel Prior from scratch as the baseline restoration method.
- Build a real, self-captured dataset with clear references across four media (haze, water, glass, plastic) at three severity levels.
- Compare DCP against classical and medium-aware alternatives on every medium.
- Build a synthetic degradation model (single-layer and stacked multi-layer) and validate it against real captures.
- Explain successes and failures using dark-channel statistics, transmission behaviour and colour-cast analysis.
- Recommend a simple, medium-aware restoration pipeline for each medium, backed by the results.

1 Data: Real, Synthetic and Existing Image Sets

Real captures are the primary evidence. Matched synthetic images (degrading our own clear references with the forward model, then comparing against the real degraded capture of the same scene) validate that model. Bulk synthetic data, generated from public depth datasets, drives severity sweeps and stacked-layer experiments that would be impractical to capture by hand.

Medium	Real (own captures)	Synthetic (approximation model)	Existing online image sets
Haze	Outdoor haze/fog shots plus a clear reference on a clear day	t = exp(-βd); A near white	O-HAZE / I-HAZE (real haze, paired GT); RESIDE SOTS (paired); RESIDE ITS / NYU Depth v2 (depth, for synthesis)
Water	Aquarium/tub with a reference shot; milk drops for turbidity levels	Per-channel t_c = exp(-β_c d), β_R > β_G > β_B; blue-green A	UIEB for real underwater images (pseudo-reference only; rely on no-reference metrics)
Glass	Fog by steam/spray, dirt by dust/smudges, on a clear pane	Near-uniform low-frequency veil (blurred noise), droplet patterns, alpha-blended smudge masks	None fit – captures and synthesis carry this medium
Plastic	Clear, translucent and aged plastic sheets over a scene	Uniform low t, tint gains, Gaussian blur	None fit – captures and synthesis carry this medium
Stacked	Water/haze scene viewed through a fogged or dirty pane	Two-layer model: I2 = J·t1t2 + A1t2(1-t1) + A2(1-t2)	—

Caveat. Synthetic haze generated with the same model DCP inverts favours DCP by construction, so the synthetic-real gap is reported explicitly rather than hidden. Synthesis parameters are calibrated on a few scenes and evaluated on held-out ones. Depth for our own scenes comes from measured distances, a planar ramp, or 2–3 hand-defined layers — no learned depth model is used.

2 Pipeline: Hand-Implemented vs Library

The rule: anything under study or central to the project's claims is implemented by hand; generic, uncontested primitives use a library. Library use for baselines should be confirmed with the instructor.

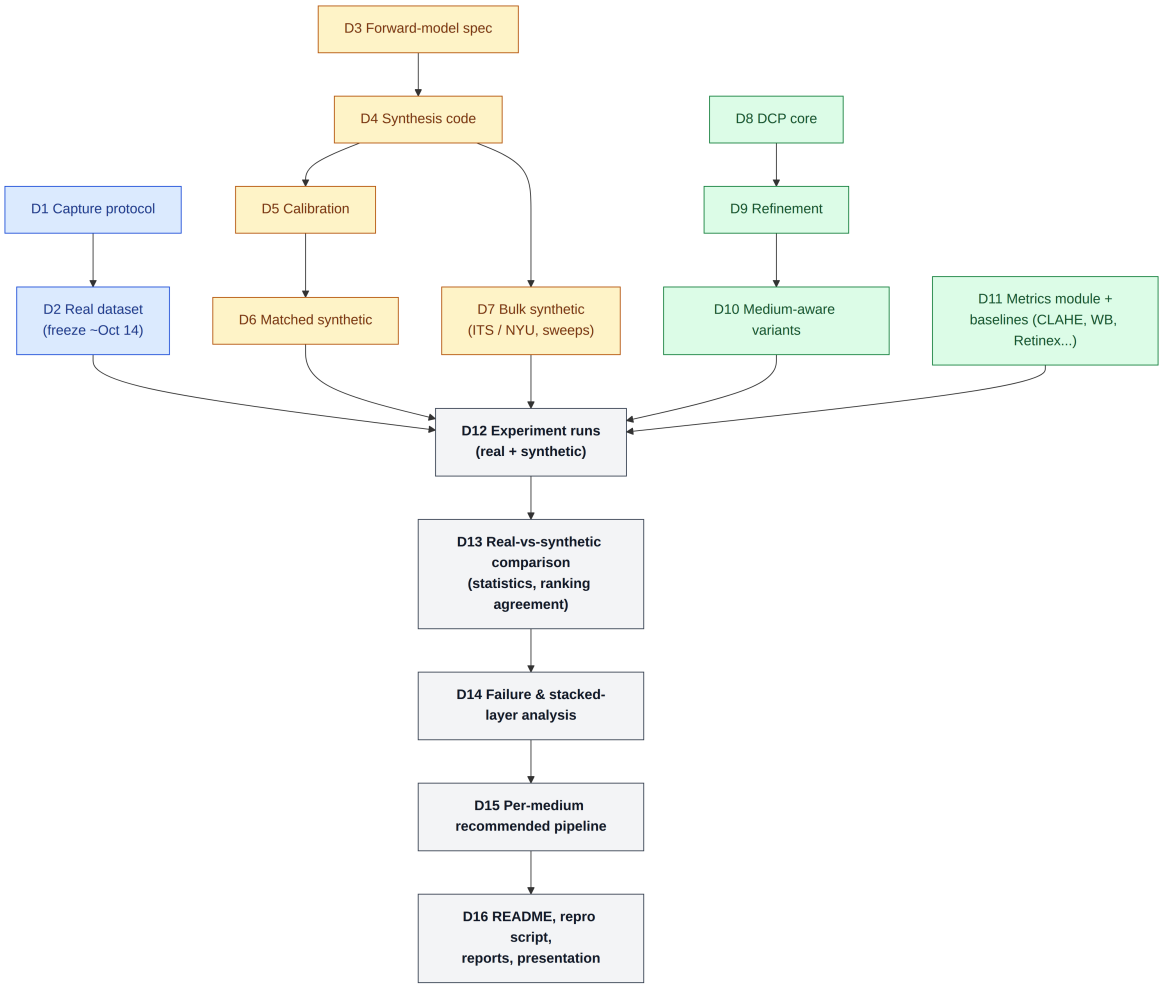
Stage	By hand	Library
Dark channel (patch-min over channels)	Yes – pipeline logic is ours	scipy.ndimage / cv2.erode for cross-check
Atmospheric light, raw transmission, recovery (t0 floor)	Yes	–
Guided filter refinement	Optional manual version	cv2.ximgproc.guidedFilter (opencv-contrib)
Soft matting (stretch goal)	–	pymatting
Medium-aware variants: UDCP/red channel, per-channel or local A, large-patch/near-global t	Yes – modifications of our DCP	–
Gray-world / simple white balance	Manual gains	cv2.xphoto as cross-check
Homomorphic filtering (glass)	Yes – log, FFT, high-pass, exp	NumPy FFT
Wiener deblur (plastic)	PSF estimation manual	skimage.restoration.wiener
CLAHE, histogram equalization, gamma	–	OpenCV
Retinex (MSR, stretch)	Yes – Gaussian-blur based	–
Median / bilateral cleanup	–	OpenCV
Forward degradation models (all media, stacked)	Yes	NumPy / SciPy for noise & blur helpers
PSNR, SSIM, CIEDE2000	–	skimage.metrics, skimage.color
UCIQE, entropy, contrast, dark-channel histograms, model-fit residual, ranking correlation	Yes – short formulas, part of the analysis	scipy.stats for Spearman/Kendall
Plots and tables	–	Matplotlib, pandas

Generalized skeleton with per-medium presets: preprocess (none / white balance) → medium-light estimate (global / per-channel / local) → dark-channel variant → transmission (raw, then guided filter) → recover → postprocess (CLAHE or contrast stretch). A config file selects the modules per medium. CAP is excluded because its coefficients were fit by supervised learning.

Medium	Variants tested
Haze	DCP, DCP + guided filter, Retinex, CLAHE
Water	White balance then DCP, UDCP, CLAHE
Glass	Large-patch/near-global t, local A, homomorphic filtering, CLAHE
Plastic	White balance then DCP, Wiener deblur, Retinex, CLAHE

3 Deliverables: Order and Dependencies

Three tracks run in parallel – real data (D1–D2), synthetic data (D3–D7) and restoration (D8–D11) – and converge at D12. From D12 onward the pipeline is strictly serial through to the final deliverables.



Deliverable dependency graph (RAG): blue = real-data track, amber = synthetic-data track, green = restoration track, grey = the serial spine from experiment runs to final deliverables.

Order of execution

Runs in parallel: D1→D2, D3→D4→D5→D6 and D4→D7, and D8→D9→D10 alongside D11, are three independent tracks until they all feed D12. Restoration code (track B) is tested on O-HAZE / RESIDE images while it waits for real and synthetic data to land.

Strictly serial: D2 (freeze) and D6/D7 → D12 → D13 → D14 → D15 → D16.

Risk assessment (RAG status)

Deliverable	Status	Why
D8–D10 DCP core & variants	Green	Well-documented algorithm, small code surface
D11 Metrics & baselines	Green	Mostly library calls
D4–D7 Synthesis	Amber	Matching real captures needs tuning
D13 Real-vs-synthetic comparison	Amber	Depends on both data tracks landing on time
D14 Stacked-layer experiment	Amber	Needs extra captures
Soft matting & Wiener tuning	Amber	Slow / fiddly – first to cut if behind
D2 Real dataset	Red	Capture logistics, lighting and tripod consistency – main schedule risk

4 Work Division

Member	Owns	Deliverables
A – Data & captures	Capture protocol, all real scenes and severity control, depth setups, dataset organisation, downloads (O-HAZE, UIEB, ITS/NYU)	D1, D2, later stacked real shots, demo images
B – Restoration & evaluation	DCP from scratch, refinement, medium-aware variants, baselines, metrics module, experiment runner	D8–D12, D14, D15
C – Synthesis & comparison	Forward models, calibration, matched & bulk synthetic data, real-vs-synthetic statistics and ranking agreement	D3–D7, D13, stacked-layer experiment
All	Understand and be able to explain the DCP core; report sections; rehearsal	D16

A few hours per week per person is realistic for this scope, provided the real dataset is frozen early (around Oct 14) so the busiest coursework weeks are not competing with capture logistics.

5 Timeline Against Course Dates

Dates	Milestone
Sep 28 – 30	Report 1: framing, pipeline diagram, capture protocol, one or two DCP outputs
Oct 1 – 14	Captures (haze & glass, then water & plastic), DCP + refinement, synthesis code. Data frozen ~Oct 14
Oct 15 – 26	Variants, baselines, metrics, calibration, first real-vs-synthetic comparison. Report 2 on Oct 26
Oct 27 – Nov 8	Ranking agreement, failure & stacked-layer analysis, severity sweeps, recommended pipelines, README
Nov 9 – 12	Rehearsal. Presentation on Nov 12 (Nov 9 is a holiday – Diwali)

Assign 3 (due Oct 8), Quiz 2 (Oct 22) and Assign 4 (due Nov 2) overlap this plan, which is why the real dataset freezes before the busiest coursework stretch.

6 Cut List and Open Items

Cut in this order if behind schedule: soft matting → Retinex → plastic stacked-layer cases → bulk severity sweeps.
Must survive: real captures, DCP plus 2–3 classical alternatives, matched synthetic data, and the ranking-agreement comparison.

To confirm with the TA / instructor:

- Whether library calls (guided filter, CLAHE, Wiener, metrics) are acceptable for baselines.
- Whether the reframing from a DCP-robustness study to a multi-method, multi-medium comparison is acceptable, since the submitted proposal was DCP-centric.
- Exact Report 1 requirements.